

Дурыгин Олег

Лекция 4. Ивентовые данные. Детекция действий по видео модальности.

План

01

Лекция:
17:00 - 18:10
70 минут

02

QA сессия:
18:10 - 18:25
15 минут

03

Перерыв:
18:25 - 18:30
5 минут

04

Семинар:
18:30 - 19:15
45 минут



У тебя есть лекция?

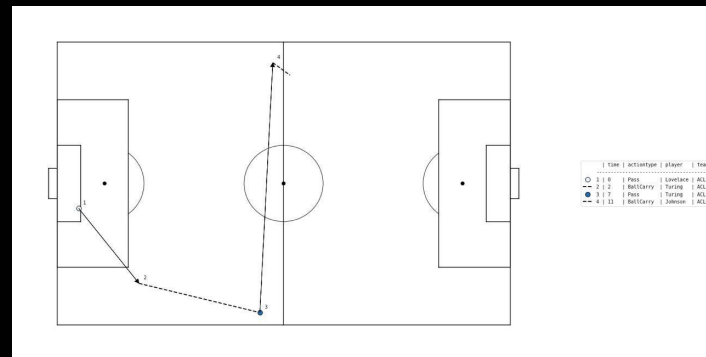
Лучше. У меня есть план лекции.

Введение + Ресурсы

Типы данных в футболе



Ивентовые данные



Ивентовые данные

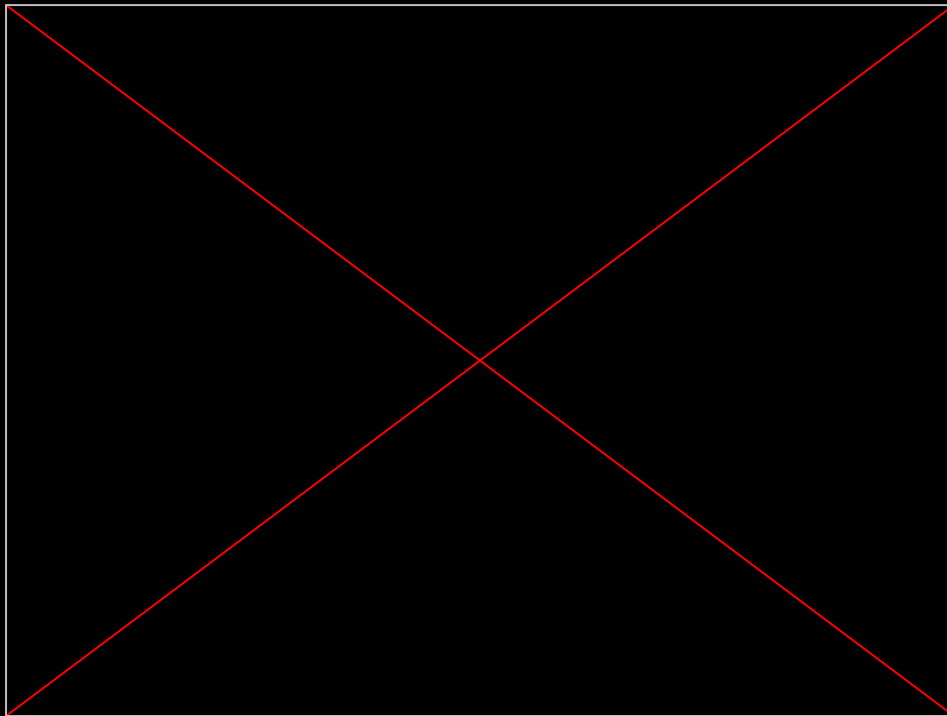
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	id	index	period	timestamp	minute	second	possession	duration	match_id	type_id	...	ball_recovery_recovery_failure	foul_won_defensive	pass_backheel	block_offensive	shot_one_on_one	off_camera	substitution_replacem
0	0d8dcd1a-710e-4b8b-b6b2-4d75e6d2d334	1	1	00:00:00	0	0	1	0.000000	19720	35	...	NaN	NaN	NaN	NaN	NaN	NaN	
1	f8823c57-870e-4b84-8fa3-0a2fabcb6524c	2	1	00:00:00	0	0	1	0.000000	19720	35	...	NaN	NaN	NaN	NaN	NaN	NaN	
2	0730284c-38f6-4f12-9a78-ebb4367708f3	3	1	00:00:00	0	0	1	0.000000	19720	18	...	NaN	NaN	NaN	NaN	NaN	NaN	
3	a21f5680-ca44-45c9-bc1e-15737d559b3b	4	1	00:00:00	0	0	1	0.000000	19720	18	...	NaN	NaN	NaN	NaN	NaN	NaN	
4	d4b3c3ae-ed78-42f1-81e8-b961ee7f281e	5	1	00:00:01.289000	0	1	2	1.504113	19720	30	...	NaN	NaN	NaN	NaN	NaN	NaN	

5 rows x 63 columns

Способы сбора ивентовых данных

Ручной сбор



Ручной сбор



Автоматический сбор

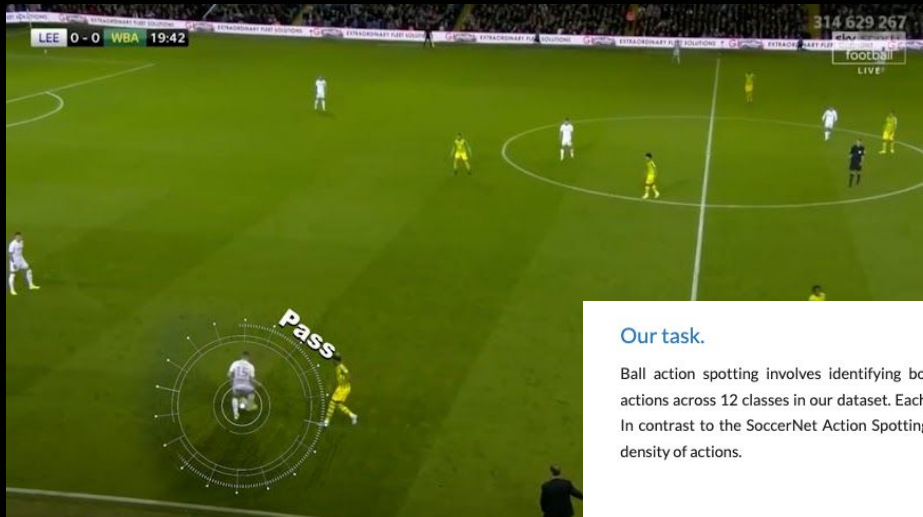
С помощью эвристик на трекинг-данных



Автоматический сбор. Бундеслига



Автоматический сбор. Computer Vision



Our task.

Ball action spotting involves identifying both the timing and type of ball-related actions across 12 classes in our dataset. Each action is marked by a single timestamp. In contrast to the SoccerNet Action Spotting Challenge, our videos feature a higher density of actions.

Our data.

The data consists of 7 videos of English Football League games, each broadcast in 720p quality. We don't provide extracted features to encourage the development of end-to-end methods. The challenge set is composed of 2 separate games. Additionally, participants are encouraged to leverage the 500 unannotated games from the Action Spotting challenge as a resource for this task.

All of our classes.

We provide annotation for all two types of soccer ball actions: {Pass, Drive}. In 2024, we extend it to 12 different classes of soccer ball actions: {Pass, Drive, Header, High Pass, Out, Cross, Throw In, Shot, Ball Player Block, Player Successful Tackle, Free Kick, Goal}

Our Metric.

Unlike the SoccerNet Action Spotting Challenge, that uses the *tight average-mAP* ([1-5] seconds tolerance), this challenge focuses on accurately spotting the ball events with the *mAP@1* (1 second tolerance).

Существующие решения

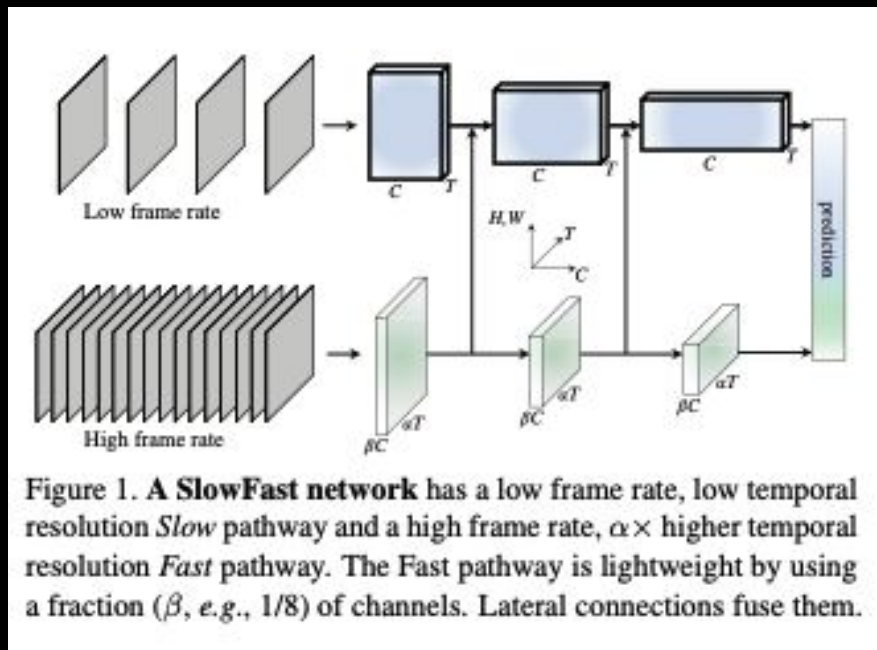
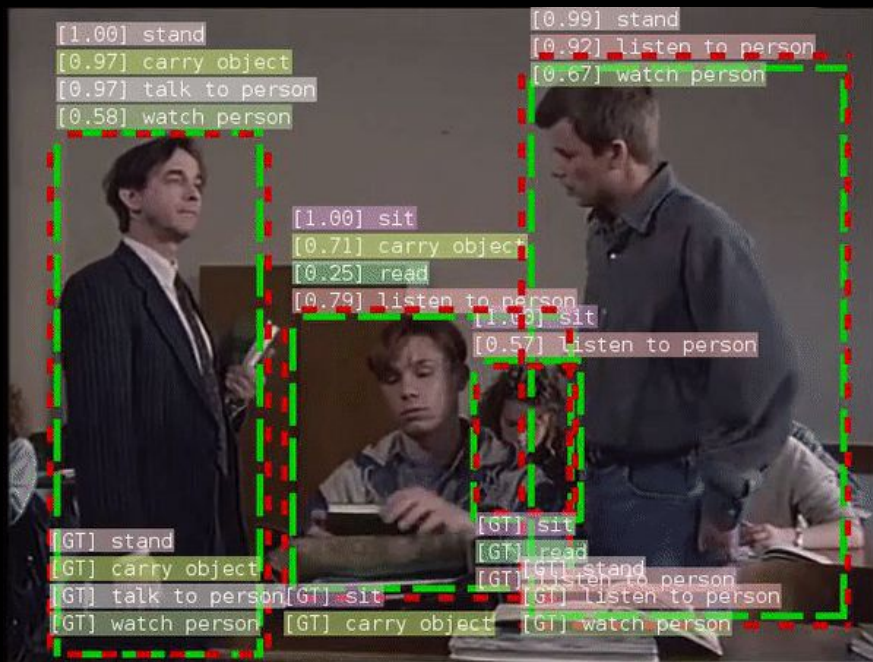
CV методы детекции действий

На данный момент можно выделить следующие базовые подходы :

- Двухэтапный подход на основе трекинга
- 3DCNN + GNN
- Трансформеры



SlowFast (2018)



A Context-Aware Loss Function (2019)

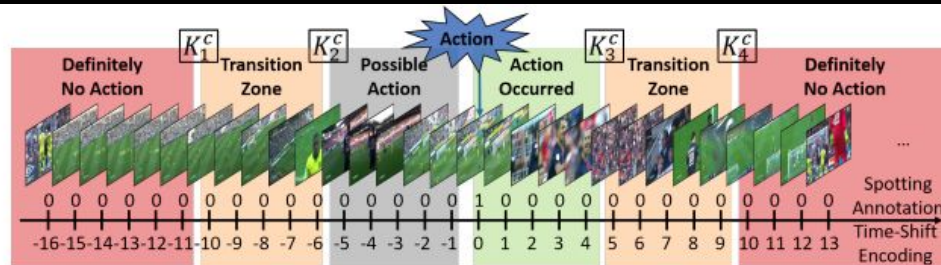
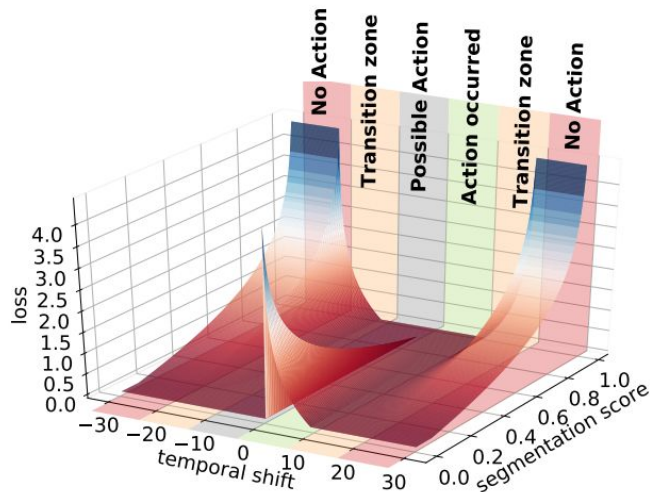
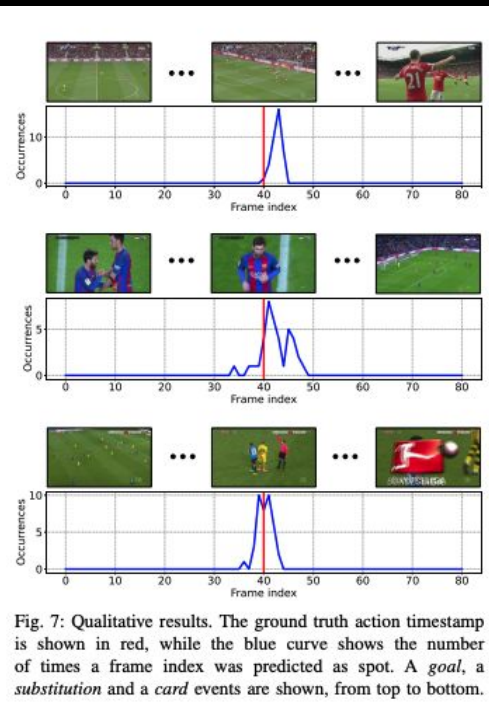
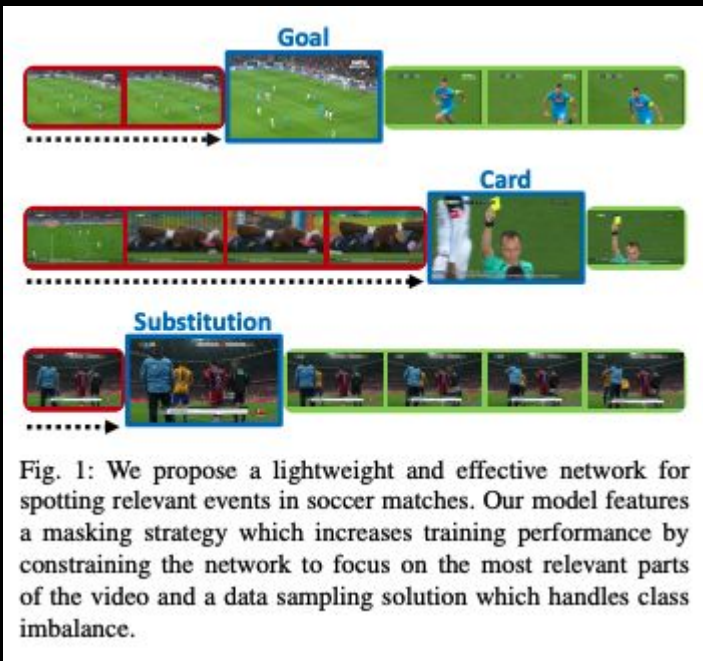
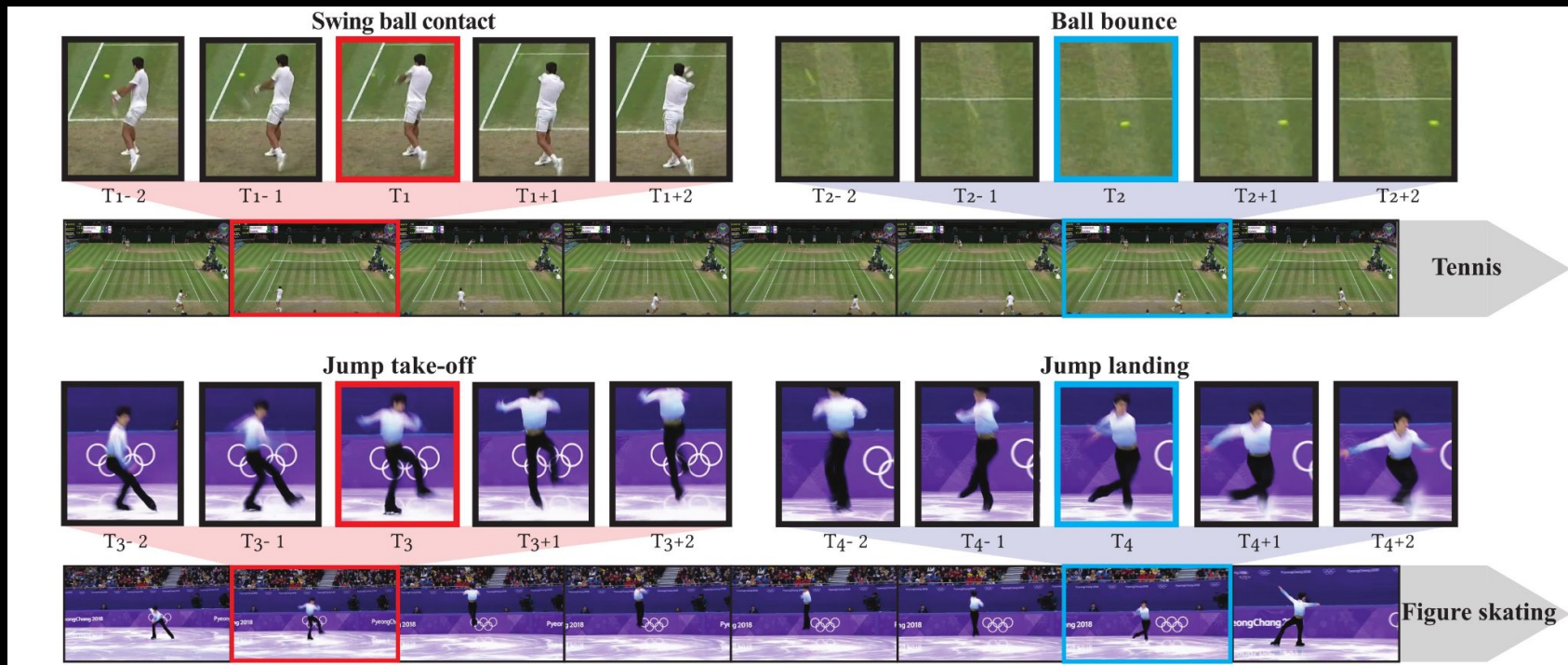


Figure 2. **Action context slicing.** We define six temporal segments around each ground-truth action spot, each of which induces a specific behavior in our context-aware loss function when training the network. **Far before** and **far after** the action, its influence is negligible, thus we train the network not to predict an action. **Just before** the action, we do not influence the network since a particular context may or may not result in an action (*i.e.* an attacking phase *may* lead to a goal). **Just after** the action, its contextual information is rich and unambiguous as the action has just occurred (*i.e.* a goal *leads* to celebrating). Hence, we train the network to predict an action. Finally, we define **transition zones** for our loss function to be smooth, in which we softly train the network not to predict an action. For each class c , the temporal segments are delimited by specific slicing parameters K_i^c and are materialized through our time-shift encoding, which contains richer temporal context information about the action than the initial binary spotting annotation.

RMS-Net (2021)



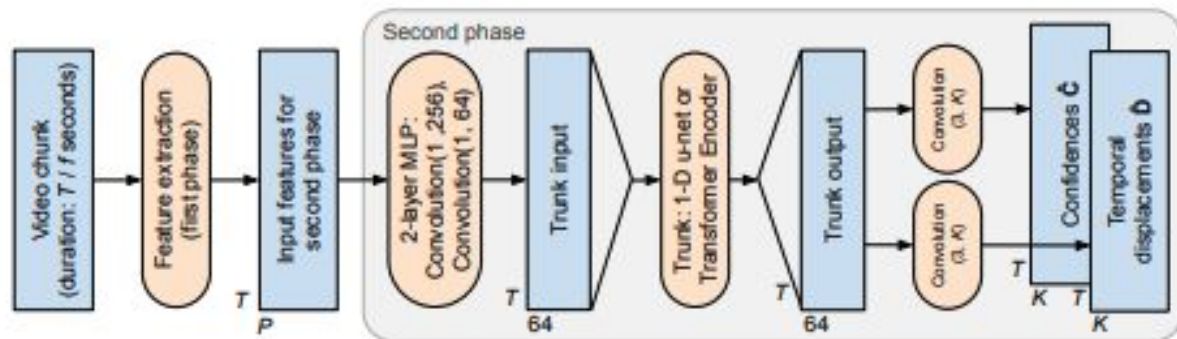
Spotting Temporally Precise (2022)



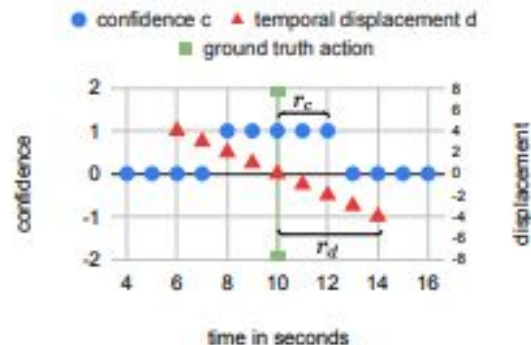
Tennis

Figure skating

Dense detection anchors (2022)



(a) Architecture overview



(b) Targets around a ground-truth action

Fig. 1: (a) Architecture overview, where $\text{Convolution}(w, n)$ denotes a convolution with kernel size w and n filters. (b) Example of target confidences $c_{*,k}$ and temporal displacements $d_{*,k}$ for a single class k , around a ground-truth action of the same class, at one anchor per second. Note temporal displacement targets are undefined outside of the radius r_d of the ground-truth action.

ASTRA (2024)

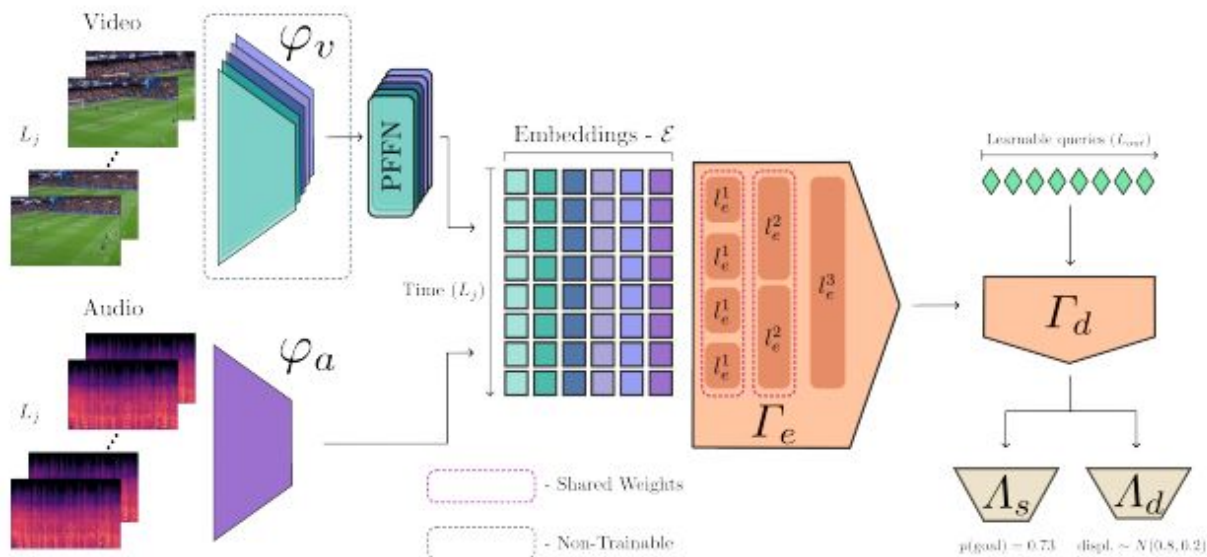
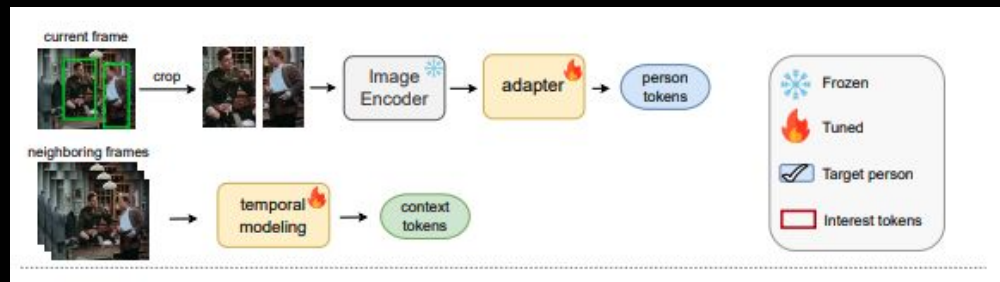
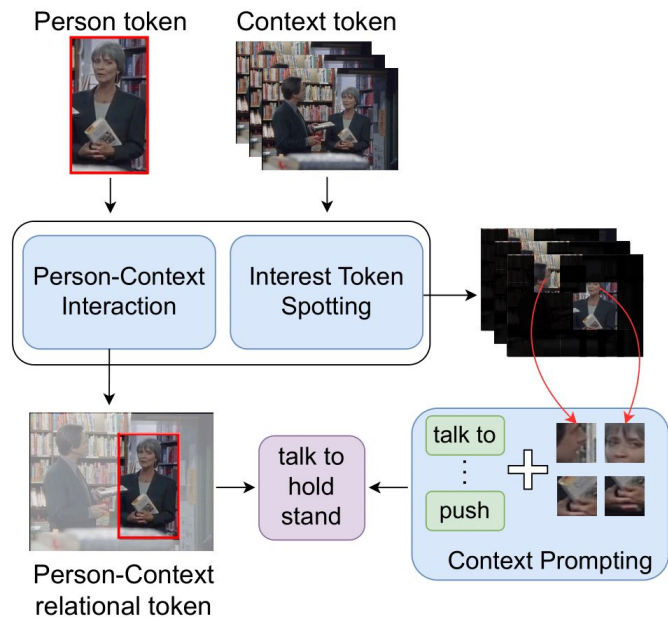


Figure 1: ASTRA (Action Spotting TRANSformer) architecture: Visual and audio embeddings from different backbones (φ_v and φ_a) are combined and processed through a Transformer encoder-decoder (Γ_e and Γ_d). The resulting embeddings are then utilized by a classification head (Λ_s) for temporal location classification, and a displacement head (Λ_d) to further refine predictions.

ST-CLIP (2024)



VideoMamba (2024)

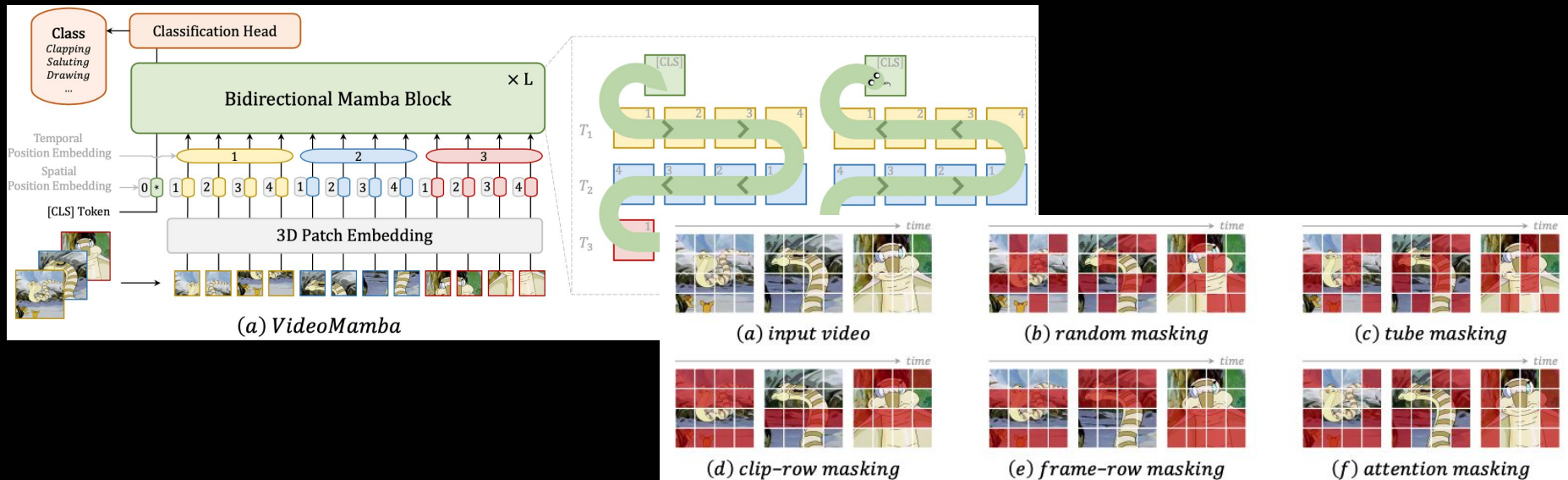


Fig. 5: Different masking strategies. Row masking, tailored for VideoMamba in light of the 1D convolution preceding SSM, enhances performance with continuous tokens. The difference between clip-row and frame-row masking is that the former masks the entire video clip, while the latter masks each frame individually.

TAAD + Graph Neural Networks (2025)

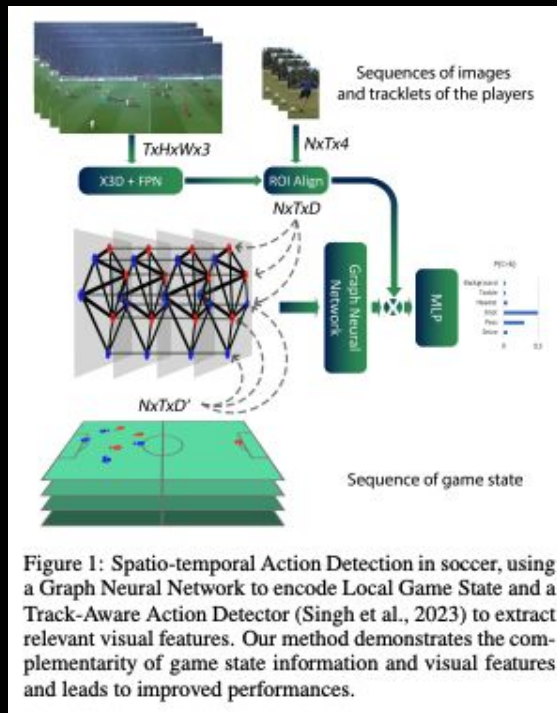


Figure 1: Spatio-temporal Action Detection in soccer, using a Graph Neural Network to encode Local Game State and a Track-Aware Action Detector (Singh et al., 2023) to extract relevant visual features. Our method demonstrates the complementarity of game state information and visual features and leads to improved performances.

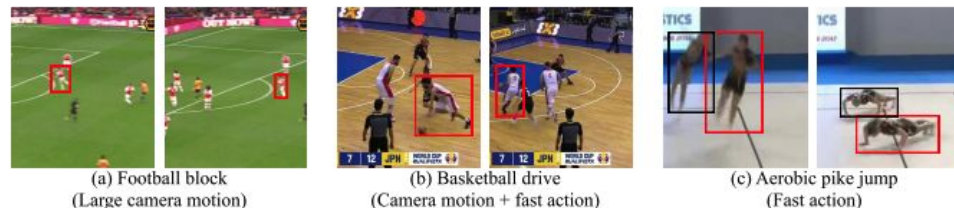


Figure 2: Reasons for large motions: (a) large camera motion (b) camera motion plus actor motion (c) static camera but super fast action. Note that, (b) shows camera zoom out and translation at the same time, and (c) shows Pike-jump action which involves jumping from standing position to air while bringing head and knee close to each other, then landing in horizontal shape on the ground, all this in close to one second. All these images contain pairs of boxes of the same actor, separated by one second in a tube with 0.0 IoU.

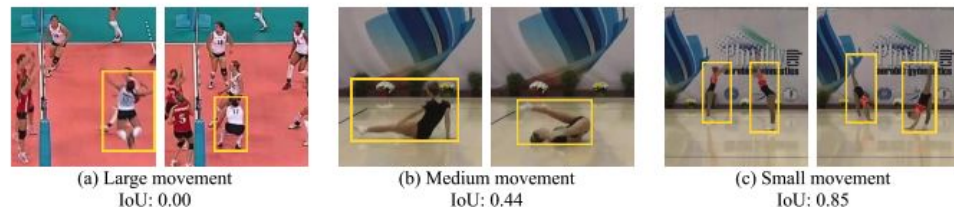
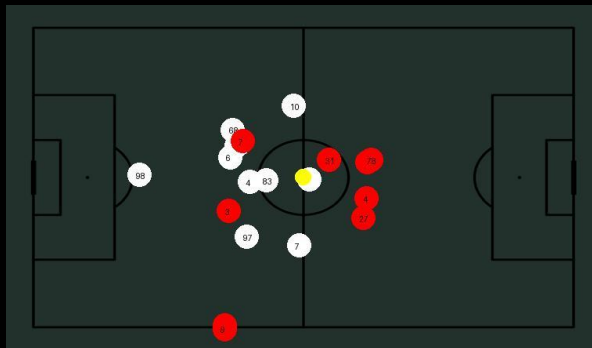


Figure 3: Varying degrees of motion observed for actors in bounding boxes with one second time window with mostly static camera: (a) large motion where actor performs spiking action, results in IoU of 0.0, meaning large-motion. (b) some pose change as resulting in 0.44 IoU, meaning medium-motion. (c) change in body pose at same location with IoU being close to 0.85, i.e. small-motion.

VLM

VLM

На 5-й минуте первого тайма, при счете 1:0 в пользу ЦСКА, мяч находится в центре поля. Игрок в белой форме контролирует мяч в центре. Большое скопление игроков наблюдается по отдалению от мяча, возможно произошла пауза в матче.



Всем спасибо за внимание!

